Chpater4 Optics & Light

Lesson2 Refraction & Snell's Law

1. A light ray of wavelength $589\ \text{nm}$ (produced by a sodium lamp) traveling through air strikes a smooth, flat slab of crown glass at an angle of $30.0°$ to the normal. Find the angle of refraction.
   Given: refractive index of a smooth, flat slab of crown glass is 1.52, $\sin 30° = 0.5$.
2. If the index of refraction for a certain glass is 1.50, and the angle of refraction is 15 degrees for a ray of light traveling from air, what is the angle of incidence? **Given:** $\sin 15° = 0.26$.
3. A coin is placed at a depth of $15\ \text{cm}$ in a beaker containing water. The refractive index of water is $4/3$. Calculate height through which the image of the coin is raised.
4. The floor of a water tank appears at a depth of $2.5\ \text{m}$. If the refractive index of water is 1.33, find the actual depth of water.
5. The depth of water in a bucket is $40\ \text{cm}$ but its bottom appears to be raised by $10\ \text{cm}$. What is the refractive index of water?
6. A small fish is swimming at a depth of $3.00\ \text{m}$ below the surface of a pond. The refractive index of water is 1.33.
   (a) What is the apparent depth of the fish as viewed from directly overhead?
   (b) If your face is $3.00\ \text{m}$ above the water surface, at what apparent distance above the surface does the fish see your face?

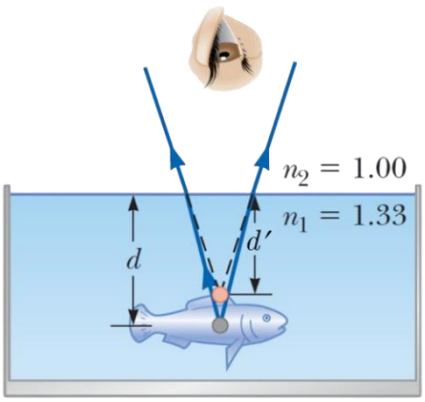


Fig. A

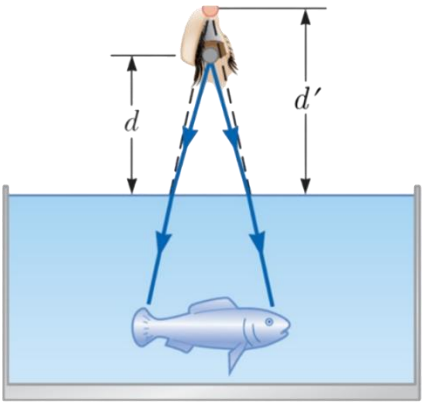
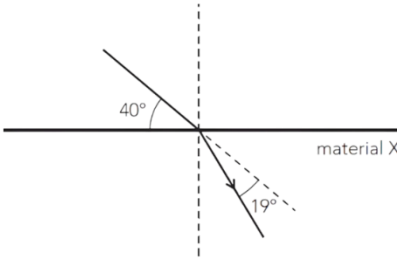


Fig. B

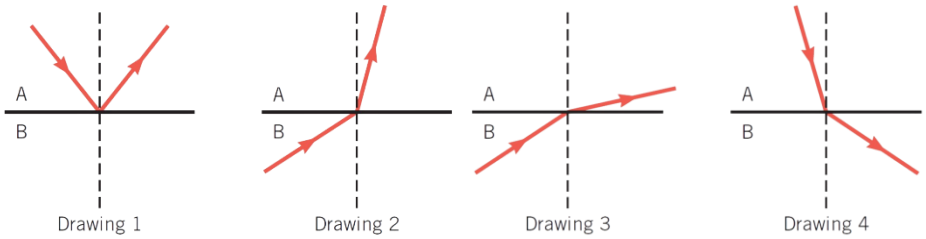
7. The refractive index of a plastic block is 1.47. What's the speed of light in the plastic?
   Given: The speed of light in air or vacuum is $3 \times 10^8\ \text{m/s}$.
8. A ray of light hits the surface of water at an angle of incidence of $30°$. It is refracted at an angle of $22°$. Find the refractive index, n, of water.
9. **Figure 13.21** shows a ray of light passing from air into an unknown material (X). The ray is deflected by $19°$.
   a Determine the angles of incidence and refraction.
   b Calculate the refractive index of material X. Give your answer to two significant figures.



Ray of light passing into air.

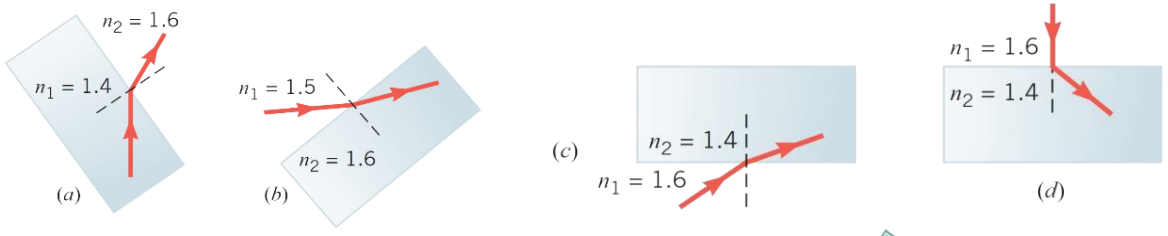
10. A ray of light enters a block of glass at an angle of incidence of $40°$. The angle of refraction in the glass is $25°$. Calculate the refractive index

11. The refractive index of material A is greater than the refractive index of material B. A ray of light is incident on the interface between these two materials in a number of ways, as the drawings illustrate. The dashed lines denote the normals to the interfaces. Which one of the drawings shows a situation that is *not* possible? **(a)** Drawing 1 **(b)** Drawing 2 **(c)** Drawing 3 **(d)** Drawing 4

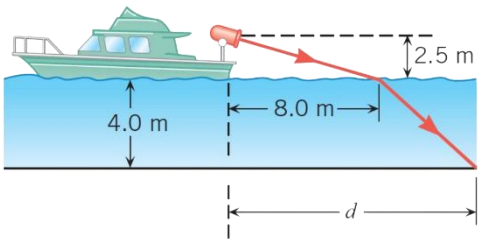


12. A light ray strikes an air/water surface at an angle of $46°$ with respect to the normal. The refractive index for water is 1.33. Find the angle of refraction when the direction of the ray is **(a)** from air to water and **(b)** from water to air.

13. The drawing shows four different situations in which a light ray is traveling from one medium into another. In some of the cases, the refraction is not shown correctly. For cases (a), (b), and (c), the angle of incidence is $55°$ for case (d), the angle of incidence is $0°$.



14. A spotlight on a boat is $2.5\ \text{m}$ above the water, and the light strikes the water at a point that is $8.0\ \text{m}$ horizontally displaced from the spotlight (see the drawing). The depth of the water is $4.0\ \text{m}$. Determine the distance d, which locates the point where the light strikes the bottom.



15. A swimmer is treading water (with her head above the water) at the surface of a pool $3.00\ \text{m}$ deep. She sees a coin on the bottom directly below. How deep does the coin appear to be?

16. A beaker has a height of $30.0\ \text{cm}$. The lower half of the beaker is filled with water, and the upper half is filled with oil $(n = 1.48)$. To a person looking down into the beaker from above, what is the apparent depth of the bottom?

17. The top of a swimming pool is at ground level. If the pool is $2.00\ \text{m}$ deep, how far below ground level does the bottom of the pool appear to be located when **(a)** the pool is completely filled with water? **(b)** When it is filled halfway with water?

18. A cubical block of ice $50.0\ \text{cm}$ on a side is placed over a speck of dust on a level floor. Find the location of the image of the speck as viewed from above. The index of refraction of ice is 1.309. Why is it harder to see dark-coloured cars at night than light-coloured ones?

19. A flint glass plate rests on the bottom of an aquarium tank. The plate is $8.00\ \text{cm}$ thick (vertical dimension) and is covered with a layer of water $12.0\ \text{cm}$ deep. Calculate the apparent thickness of the plate as viewed from straight above the water.

(a) Define the terms: angle of incidence, angle of reflection, and the normal ray.

(b) What is the relationship between the angle of incidence and the angle of reflection?

20. For the following angles, find the sine of

a) $30°$ **b)** $60°$ **c)** $45°$ **d)** $12.6°$ **e)** $74.4°$ **f)** $0°$ **g)** 9

21. For the following inverse sine values $(\sin^{-1})$, find the corresponding angle.

a) 0.342 **b)** 0.643 **c)** 0.700 **d)** 0.333 **e)** 1.00

22. Calculate the speed of light in a material with a refractive index of 0.90. Comment.

23. Sketch a light ray passing through a rectangular piece of glass. The exiting ray should be parallel to the incident ray. Draw the wavefronts.

24. Calculate the index of refraction of a diamond if the speed of light in a diamond is $1.24 \times 10^8\ \text{m/s}$.

25. Calculate the speed of light in a hypothetical material you have discovered and named in honour of yourself. Its refractive index is 0.90.

26. Calculate the speed of light for the following mediums:

a) Water $(n = 1.33)$ **b)** Diamond $(n = 2.42)$ **c)** Plexiglass $(n = 1.51)$

d) Crown glass $(n = 1.52)$ **e)** Ice $(n = 1.30)$

27. Calculate the refractive index for a substance if the speed of light in the medium is

a) $2.1 \times 10^8\ \text{m/s}$. **b)** $1.5 \times 10^8\ \text{m/s}$. **c)** $0.79c$.

28. Draw diagrams to illustrate refraction for a ray of light on:

(a) Glass - air boundary **(b)** Water - air-glass boundaries **(c)** Water - glass boundary

29. Table 6.3 shows the angles of incidence i and the angles of refraction, r, when light passes from air to glass. Complete the table and draw a graph of sin r (y-axis) against sin i (x-axis). From the graph determine the refractive index of glass.

Table 6.3

$i°$	$r°$	$\sin i$	$\sin r$
15	10		
30	19		
45	28		
60	35		

30. A ray of light passes from a liquid to air. Calculate the critical angle for the liquidair interface, if the velocity of light in the liquid is $2.4 \times 10^8\ \text{m/s}$, while in air is $3.0 \times 10^8\ \text{m/s}$.